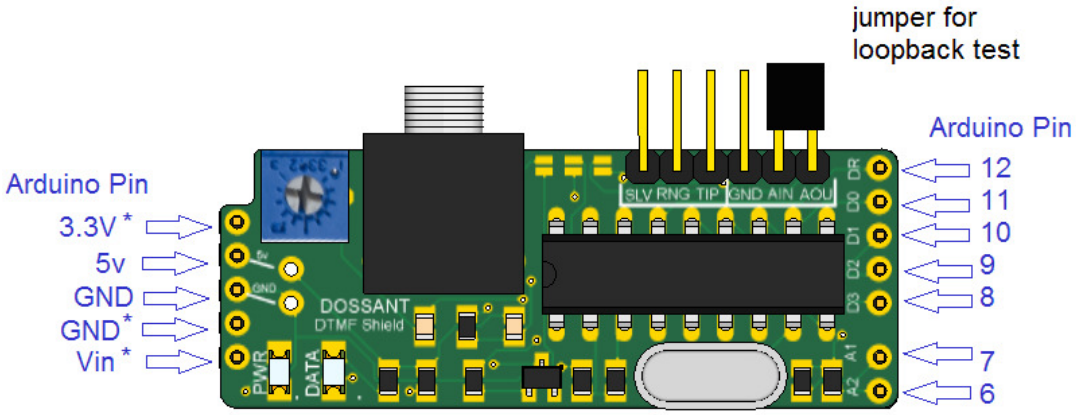
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Quick Start: Loopback test

This test is to verify the DTMF Module is working properly. A DTMF codes are generated and fed back into the audio input. Input is decoded and shown on Serial Monitor.

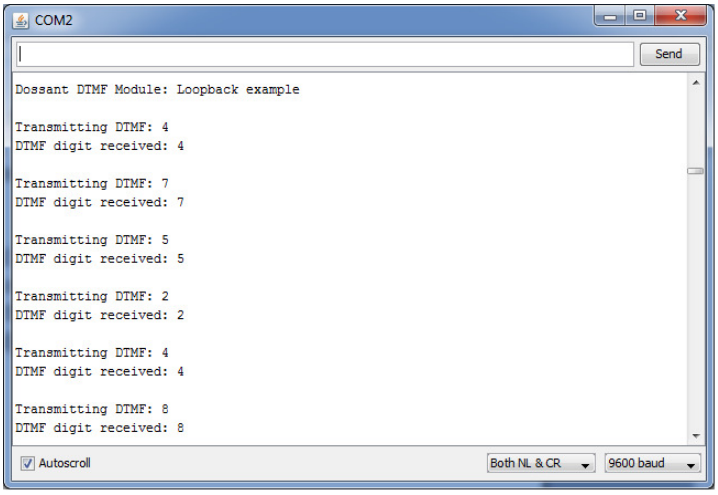
Steps:

1. Upload the loopback Sketch into your Arduino Uno board (see appendix A)
2. Power down your Arduino board
3. Remove any device from audio jack.
4. Insert jumper between AIN and AOU (see image)
5. Plug the DTMF module into Arduino. Note correct pin alignment



* pin not connected, used
just for mechanical support

6. That's it! Open Serial Monitor to confirm encoding and decoding of DTMF signals is working. Also, the DATA LED will blink to indicate tone has been decoded.



If the loopback test does not work, please adjust the gain using the blue trim pot with a screwdriver.

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Introduction

1.1 Overview

The DTMF Module converts DTMF or Touch Tone signals into a format that microcontrollers (uC) or Arduino boards can understand. DTMF tones are analog signals that Arduino and most uC cannot read directly. This module converts those signals into a digital format that uC can easily read.

The DTMF Module is specifically designed to plug right into the Arduino Uno R3, but it can also be used with a variety of microcontrollers.

Figure 1.1: DTMF Module board

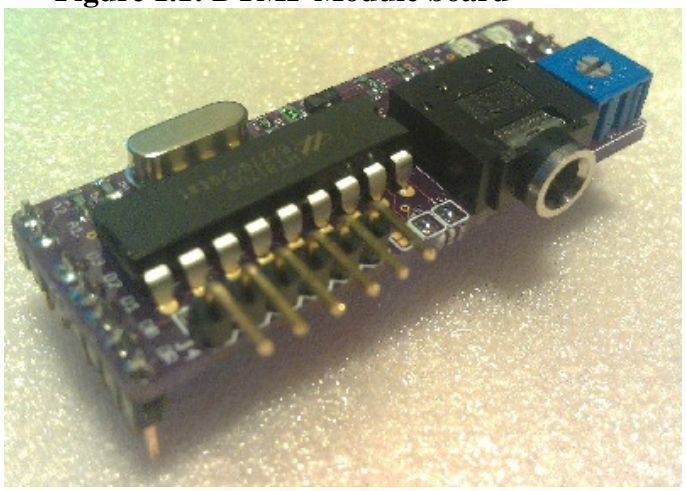
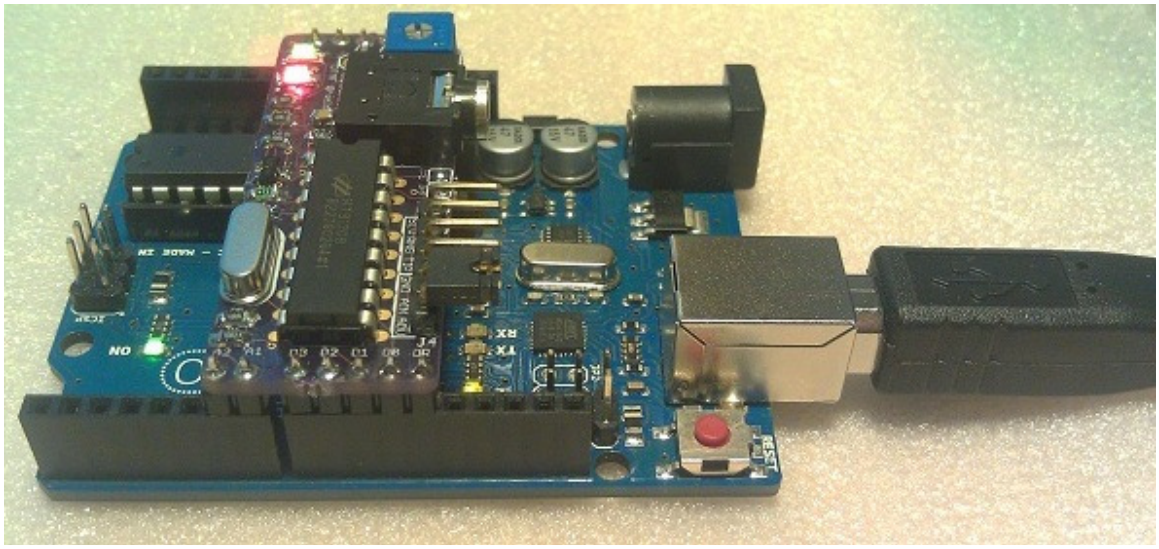


Figure 1.2:DTMF Module board setup

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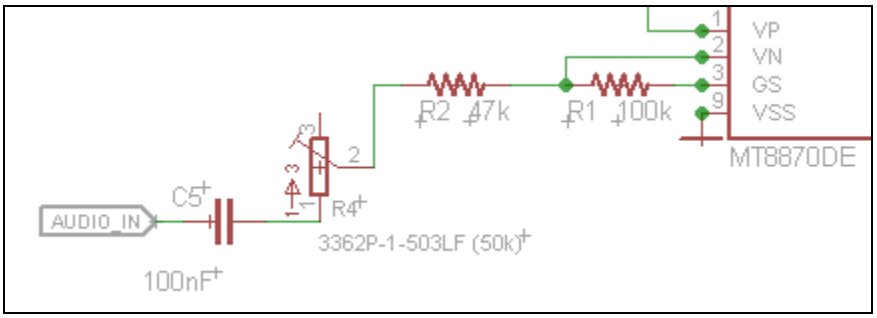


2.1 Functional Overview

DTMF decoding

The incoming audio signal goes through gain resistors and is then routed to the DTMF Integrated Circuit. The IC input stage is an inverting amplifier where the gain is:

$$G = - R1 / (R2 + R4)$$



When the potentiometer is at minimum (clockwise), R4=0 the the gain is:

Gain = - *100k/(47k+ 0)* = **2.12**

When the potentiometer is at maximum (clockwise) R4=50k and Gain is:

Gain = - *100k/(47k+ 50k)* = **1.03**

The input signal range for the DTMF IC is from 27.5mVrms to 869mVrms. Signals outside that range will not be properly decoded. So if you use let's say a cell phone to inject signals into this board, you may need to lower or raise its volume to get into the proper voltage range of the DTMF IC.

The DTMF decoder IC is the Holtek HT9170B DTMF Receiver IC.
Datasheet: http://www.holtek.com/pdf/comm/9170v111.pdf

Early boards use Zarlink MT8870DE. Both chips are identical.

DTMF encoding and playback

Pins A1 and A2 are optional if you want to generate DTMF signals and transmit them. With some simple code your Arduino can generate DTMF signals. The method I tested is using a modified Tone library where you can generate two tones at the same time, one tone per output pin. See Appendix A for sample code.

2.2 Pinout

Arduino Pin	DTMF board		Arduino Pin	DTMF board	Reference
3.3v	Not used		12	Data ready	DR
5v out	power in		11	Data bit 0	D0
Gnd	ground		10	Data bit 1	D1
Gnd	Not used		9	Data bit 2	D2
Vin	Not used		8	Data bit 3	D3
			7	Audio out 1	A1
			6	Audio out 2	A2



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Appendix A: Loopback Sketch

Download Tone library from:
https://code.google.com/p/rogue-code/downloads/detail?name=Arduino-Library-Tone.zip&can=2&q=

Note that library version V0006 has a bug. See here how to fix:

https://code.google.com/p/rogue-code/issues/detail?id=6&colspec=ID%20Type%20Status%20Priority%20Reporter%20Owner%20Summary

Example Code: dtmf_plugin_loopback.ino

```
/*
  dtmf_plugin_loopback
  this is a demo code for use with DTMF Module available at


  Product page:

  This code sends a random dtmf code on pins 11 and 12 every second,
  and receive dtmf digits via pins 3, 4, 5 and 6 upon a rising edge
interrupt on pin 2

  for loopback, add jumper to J4 pins AIN and OUT

 created 2-12-2013 by Igor Ramos

 */
#include <Tone.h>

Tone freq1;
Tone freq2;

const int DTMF_freq1[] = { 1336, 1209, 1336, 1477, 1209, 1336, 1477,
1209, 1336, 1477 };
const int DTMF_freq2[] = {  941,  697,  697,  697,  770,  770,  770,
852,  852,  852 };

int dr = 12; //dr goes high when data ready
int d0 = 11;
int d1 = 10;
int d2 = 9;
int d3 = 8;
int a1 = 6;
int a2 = 7;
```

```
char data_ready=0, data[4] = {0,0,0,0};
char data_ready_status=0;
char dr_serviced=0;
char tx_in_progress=0;


void setup() {

     Serial.begin(9600);

     Serial.println("Dossant DTMF Module: Loopback example");

  // initialize the digital pin as an output.
    pinMode(d0, INPUT);
    pinMode(d1, INPUT);
    pinMode(d2, INPUT);
    pinMode(d3, INPUT);
    pinMode(dr, INPUT);

    freq1.begin(a1);
    freq2.begin(a2);

}



// the loop routine runs over and over again forever:
void loop() {
static unsigned long int time, task, tx_time;
char tmp;

    time = millis();

    data_ready_status = digitalRead(dr);

    if (time - task > 1000) // send random char every 500ms
    {
       task = time;

       if (tx_in_progress)
         Serial.println("Rx Timeout");

        char randNumber = random(9);
        Serial.print("\n\rTransmitting DTMF: ");
        Serial.println(randNumber,DEC);

        playDTMF(randNumber, 500);
        tx_in_progress =1;
    }

    if (data_ready_status)
```

```
    {
      if (!dr_serviced)
      {
        //read data pins
        data[0] = digitalRead(d0);
        data[1] = digitalRead(d1);
        data[2] = digitalRead(d2);
        data[3] = digitalRead(d3);

        tmp = dtmf_digit(); //assemble dtmf digit;
        Serial.print("DTMF digit received: ");
        Serial.println(tmp,DEC);

        dr_serviced =1; //clear received flag
        tx_in_progress =0;    //clear  tx in progress flag
      }

    }
    else
      dr_serviced =0;


}



// ----DTMF receive functions ----
char dtmf_digit ( void  )  //assemble the bits into a digit
{
  char dtmf_digit;

// dtmf digit is decoded per Page 5 Table 1 of chip datasheet
// http://www.zarlink.com/zarlink/mt8870d-datasheet-oct2006.pdf

  dtmf_digit = 8* data[3] +  4* data[2] + 2* data[1] + data[0] ;

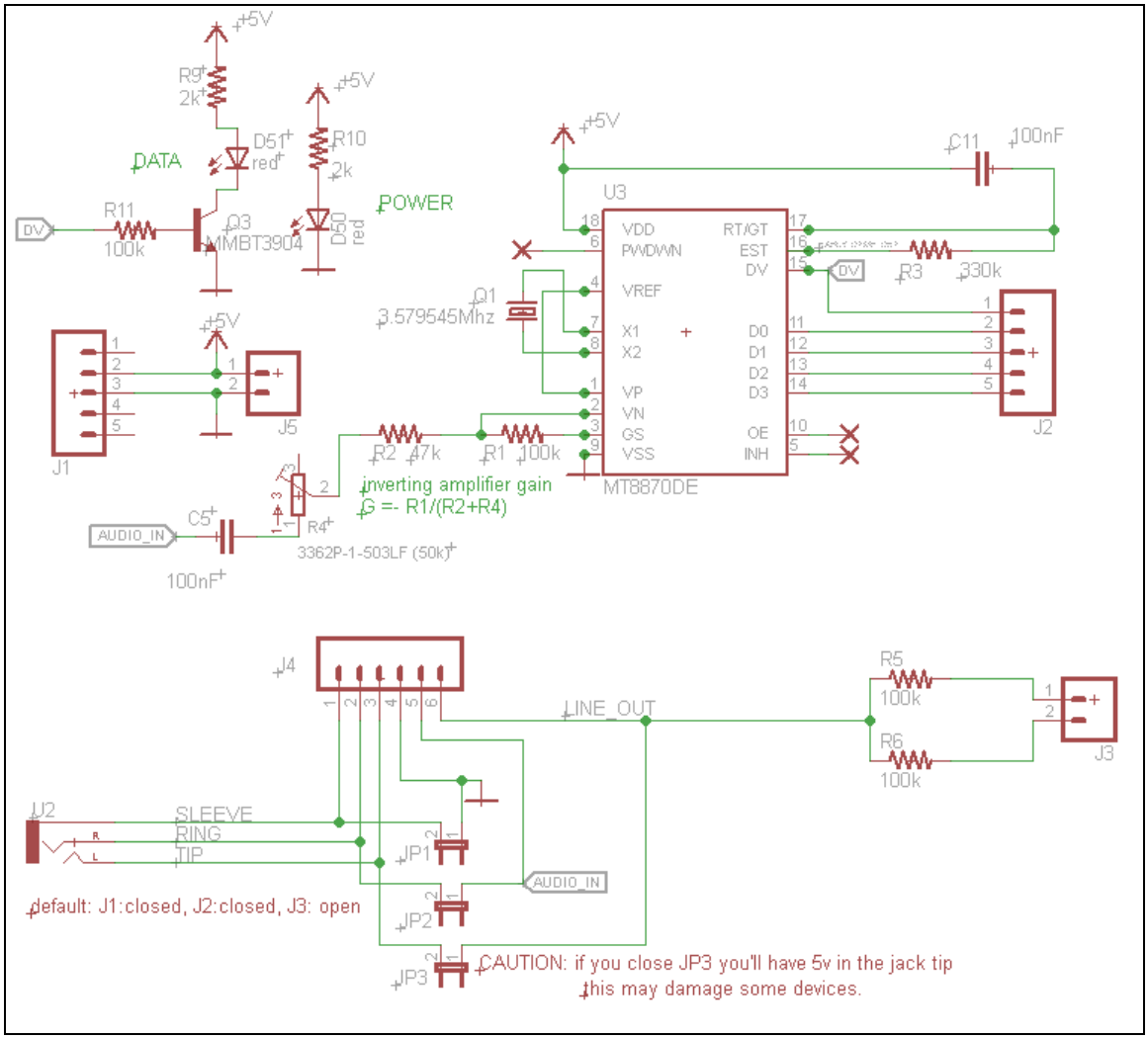
  if (dtmf_digit==10)
    dtmf_digit =0;

  return dtmf_digit;
}

// ----DTMF Send functions ----
void playDTMF(uint8_t number, long duration)
{
  freq1.play(DTMF_freq1[number], duration);
  freq2.play(DTMF_freq2[number], duration);
}
 //end of code
```

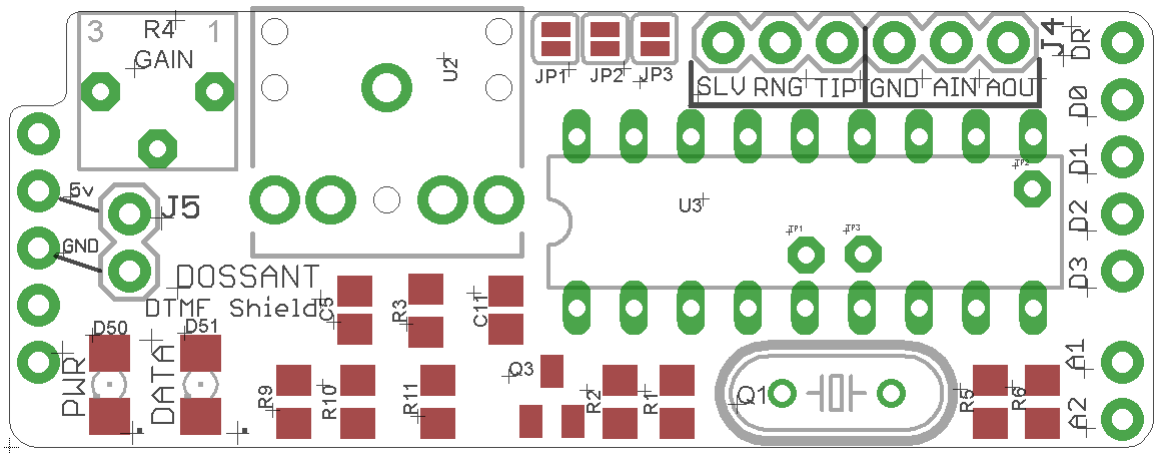
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Appendix B: Board documentation
Schematics

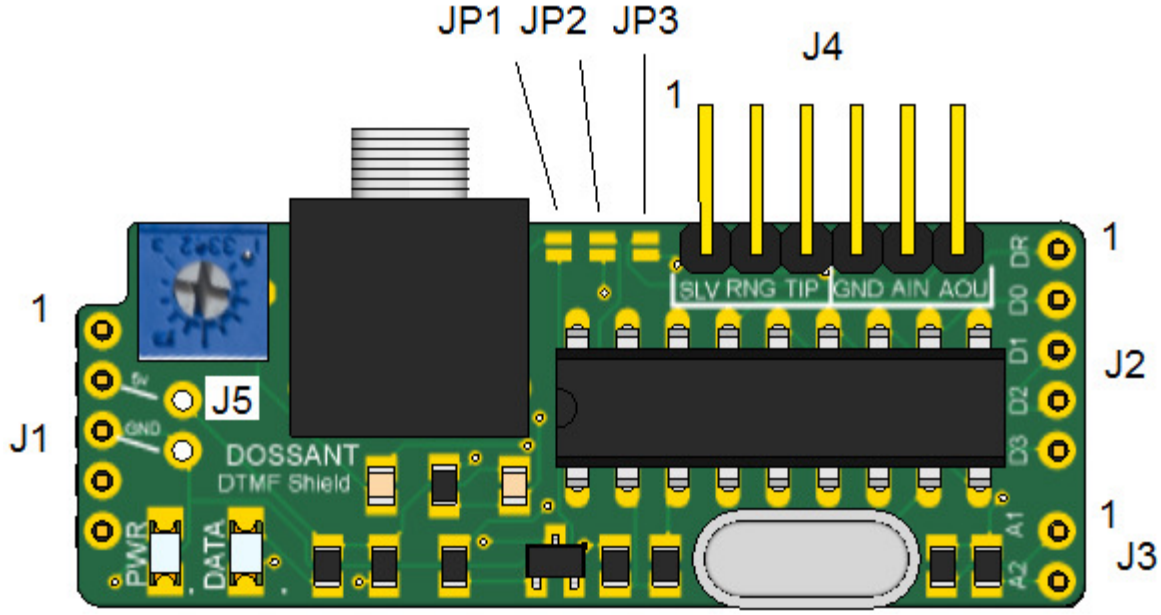


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Circuit Board component locations:



Header and Jumper locations:



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Document change history:

Version	date	By	changes
1	4/28/13	ISR	First version
1.1	4/28/13	ISR	Updated DTMF IC info. Revised Quick Start steps